# First Implementation Iteration: One-Time Classroom Allocation

## with Availability Rule

Software Design / Iteration Report

Name: Najib Attar
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## 1. Iteration Goal

The goal of this first implementation iteration is to validate the core allocation workflow by implementing the simplest possible scenario: a one-time classroom request evaluated using a single rule (AvailabilityRule). This iteration focuses on verifying that a request can be processed, evaluated, and either approved or rejected based on time-slot availability.

## 2. Scenario

A user submits a one-time classroom request for a given time slot. The system checks whether the classroom is available in that time slot. If the classroom is available, an allocation is created. Otherwise, the request is rejected.

## 3. Modules and Their Tasks

### A. Request Management Module

- Receive the request from the user
- Create and store `OneTimeRequest`

### B. Resource Management Module

- Manage classroom information
- Provide classroom details when needed

### C. Rule Evaluation Module

- Apply `AvailabilityRule`
- Determine whether the request satisfies the availability rule.

### D. Allocation Module

- Create `Allocation` when the rule passes
- Update request status

## 4. Simplified Class Diagram

Figure 1 shows the simplified class diagram for the first implementation iteration.

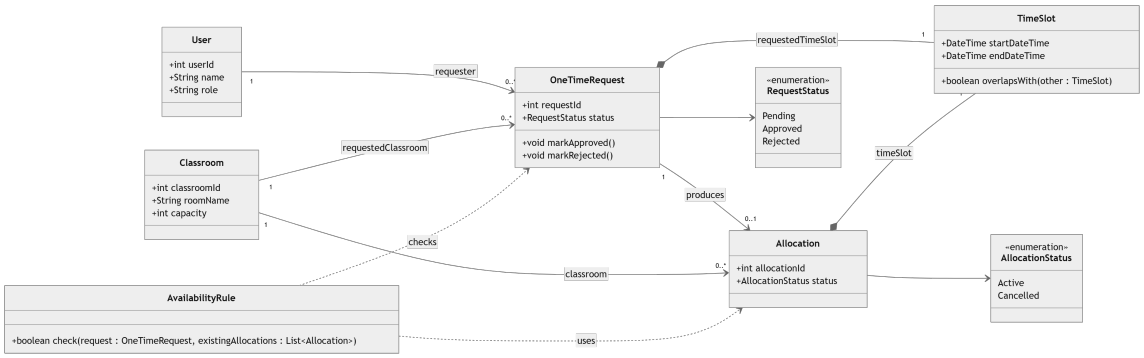


Figure 1: Simplified Class Diagram

## 5. Workflow Diagram

Figure 2 shows the workflow of the one-time classroom allocation process.

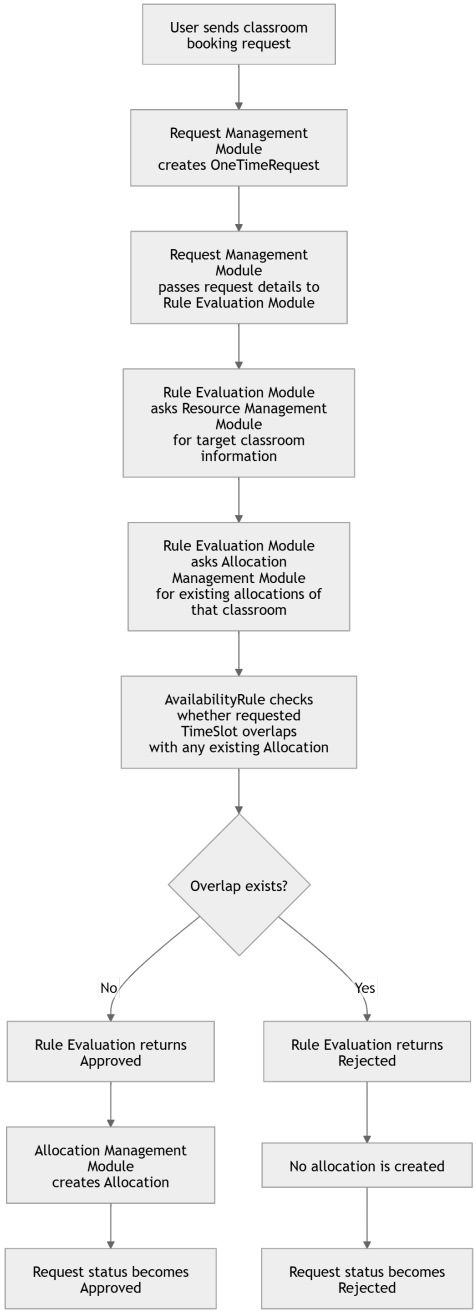


Figure 2: Workflow Diagram

## 6. Iteration Overview

| Iteration | Request Type | Resource Type | Rules |
|---|---|---|---|
| Iteration 1 | OneTimeRequest | Classroom | AvailabilityRule |

Table 1: Iteration Structure Overview

## 7. Prototype Implementation

To validate the design of this iteration, a small prototype was implemented in C++. The prototype simulates the request evaluation process by creating example classroom allocations and processing new requests. The system checks whether the requested time slot overlaps with existing allocations using the AvailabilityRule.

## 8. Summary

This first implementation iteration focuses on validating the core allocation workflow using the simplest possible system configuration. A one-time classroom request is submitted, evaluated using the AvailabilityRule, and either approved or rejected depending on whether the requested time slot conflicts with existing allocations.

This iteration establishes the basic architecture of the system and verifies the interaction between the main modules: request management, rule evaluation, resource management, and allocation management. Future iterations will extend this prototype by introducing additional rules, request types, and resource constraints.